UNIFORM ASYMPTOTIC PHASE FOR HERMITIAN–YANG–MILLS FLOW NEAR A POLYSTABLE BUNDLE

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Abstract. Near a fixed locally free polystable bundle on a compact Kähler manifold, we construct an exact finite-dimensional invariant reduction of Hermitian–Yang–Mills flow with differentiable asymptotic phase. If the harmonic deformation coefficients remain in a prescribed neighborhood and the initial metric has sufficiently small normal displacement, a selected reduced trajectory reconstructs an exact flow on the same holomorphic bundle. The relative metric error decays exponentially in a fixed Hölder norm, with constants controlled by the complementary elliptic gap and independent of the smallest positive finite-dimensional eigenvalues, their ratios, and stabilizer changes. The phase depends continuously differentiably on the initial metric, even when the reconstructing complex gauges are unbounded. The construction retains arbitrary multiplicities and the complete convergent Kuranishi obstruction. It combines exact noncommuting metric coordinates with a parameter-dependent invariant graph and a group-valued phase estimate. An obstructed repeated-factor bundle on a smooth bicubic threefold supplies noncommuting integrable coefficients with explicitly separated finite spectral scales.

1. Introduction

The Hermitian–Yang–Mills flow relates stability of a holomorphic bundle to the geometry of its Hermitian metrics. In a deformation slice near a polystable bundle, the flow has finitely many neutral directions and an elliptic complement with a positive spectral gap. The neutral directions contain the deformation and automorphism data. They may acquire arbitrarily small positive scales after deformation, and their stabilizers may change. The question considered here is whether an exact finite-dimensional trajectory can be selected from the initial metric so that the remaining metric error decays at the rate of the fixed elliptic complement.

There are two distinctions in this question. First, the selected trajectory must solve the corrected finite equation on a parabolically invariant graph. Its leading moment-map approximation need not give a uniformly accurate comparison for all time. Second, harmonic deformation coordinates alone do not determine a metric on a fixed holomorphic bundle. At the undeformed point they are zero for an entire family of parallel stationary metrics. The reconstructing complex automorphism must therefore be retained. We prove phase selection in that group, with estimates in a right-invariant metric, and then obtain an exponentially vanishing relative error between actual HYM metrics.

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1.1. **Geometric data and the local regime.** Let (X, ω_0) be a compact connected Kähler manifold, and let

$$E_0 = \bigoplus_{i=1}^{\ell} F_i \otimes W_i \tag{1}$$

be a locally free polystable holomorphic bundle. The stable factors F_i are mutually non-isomorphic and have the same slope at ω_0; the finite-dimensional multiplicity spaces W_i are arbitrary. Fix a direct-sum Hermitian–Einstein metric. Let ω_p be a smooth finite-dimensional family of nearby Kähler forms, with $p = 0$ corresponding to ω_0. The parameter is constant along every flow. Section 2 constructs smooth factor HE metrics $h_i(p)$. We write $h_p = \bigoplus_i h_i(p) \otimes I$, and fix its HE determinant metric throughout the flow. The factor HE constants are $\kappa_i(p)$; the determinant-normalized constant for E_0 is $\kappa_E(p)$.

Put $r_i = \operatorname{rk} F_i$ and let

$$G = \left\{ (g_i) \in \prod_i \mathrm{GL}(W_i) : \prod_i \det(g_i)^{r_i} = 1 \right\}^0, \qquad K = G \cap \prod_i \mathrm{U}(W_i). \tag{2}$$

These are parallel automorphisms of the split bundle. Write $\mathfrak{a} = i\mathfrak{k}$ for the Hermitian real subspace of $\mathfrak{g}$. It has dimension $\sum_i (\dim W_i)^2 - 1$, independent of the stabilizer of a deformation. Let P_p be the $L^2(h_p, \omega_p)$ projection from trace-free Hermitian endomorphisms onto $\mathfrak{a}$, and let $Q_p = I - P_p$.

Let $d = \bar\partial_{E_0}$. Smooth equivariant identifications of its harmonic spaces give a fixed complex vector space V, with harmonic inclusion $\iota_p : V \longrightarrow \Omega^{0,1}(\operatorname{End}_0 E_0)$. The canonical convergent Kuranishi reconstruction and obstruction are

$$\begin{aligned} a_p(x) &= \iota_p x - d_p^* G_{2,p}\big(a_p(x) \wedge a_p(x)\big), \\ \kappa_p(x) &= H_{2,p}\big(a_p(x) \wedge a_p(x)\big), \qquad Z_p = \kappa_p^{-1}(0). \end{aligned} \tag{3}$$

Here $H_{2,p}$ and $G_{2,p}$ are the harmonic projection and Green operator in degree two. We retain the full convergent equations. The notation gx denotes the induced representation of G, and $C_x\xi = [\xi, x]$ its infinitesimal action. Unless an inclusion is needed, we also write x for $\iota_p x$.

For nonintegral $s > 0$, h^s denotes the closure of smooth sections in the C^s norm. All section norms use the split connection and metric; the identifications in Section 2 make them uniformly equivalent for small p. On G fix a right-invariant Riemannian metric whose inner product at the identity is $\operatorname{Ad} K$ invariant, and denote its distance by d_R.

On a fixed holomorphic bundle $d_{x_b} = d + a_p(x_b)$, where $x_b \in Z_p$, our coordinates for the metric are

$$H = g^*(h_p e^u), \qquad x = g x_b, \qquad u \in Q_p h^{17/8}(\operatorname{Herm}_0 E_0). \tag{4}$$

The exact evolution of these coordinates is derived in Proposition 2.2. The coefficient x may stay small while g and g^{-1} become unbounded.

Definition 1.1. For a coefficient radius r, a solution is *confined* if its exact sliced harmonic coefficient satisfies $|x(s)| \le r$ throughout its maximal interval in the domain of the geometric coordinate equations. The normal coordinate is initially small. There is no bound on the reconstructing gauge.

Confinement is an assumption on the actual coefficient trajectory. The theorem below proves continuation in the geometric domain under this assumption. It does not assert confinement for every nearby metric.

1.2. **Main theorem.** Use the geometric L^2 inner products on V and $\mathfrak{a}$ to define the Riesz representative of the quadratic moment map by

$$\langle \mu_p^\sharp(x), \xi\rangle_{\mathfrak{a},p} = \mathrm{Re}\langle C_x\xi, x\rangle_{V,p}. \tag{5}$$

This convention has no factor $1/2$. Define the central slope difference by

$$b_p|_{F_i\otimes W_i} = (\kappa_E(p) - \kappa_i(p))I. \tag{6}$$

It belongs to $\mathfrak{a}$ and $b_0 = 0$.

Theorem 1.2. *For every prescribed finite $m \geq 6$, there are a parameter neighborhood, a radius $r > 0$, a chosen jointly C^m, K-equivariant map*

$$\Psi(p,\cdot) : B_{4r}(V) \longrightarrow Q_p h^{19/8}(\mathrm{Herm}_0 E_0),$$

and positive constants δ, C, c with the following properties.

(i) Invariant graph. *The graph $u = \Psi(p,x)$ is locally invariant for the exact sliced HYM equation, and $\Psi(p,0) = D_x\Psi(p,0) = 0$. Its exact finite evolution is*

$$\dot{x} = C_x\beta_0(p,x), \qquad \dot{g} = \beta_0(p,x)g, \tag{7}$$

where $\beta_0(p,x) \in \mathfrak{a}$ and

$$\beta_0(p,x) = b_p - \mu_p^\sharp(x) + E_p(x), \qquad |D_x^j E_p(x)| \leq C|x|^{4-j} \quad (0 \leq j \leq 2). \tag{8}$$

The leading coefficient field is $-\nabla_V\big(\frac{1}{4}\|\mu_p^\sharp - b_p\|^2_{\mathfrak{a},p}\big)$.

(ii) Asymptotic phase and metric tracking. *Fix $x_b \in Z_p \cap B_r(V)$ and initial data*

$$g_0 \in G, \quad x_0 = g_0 x_b \in B_r(V), \quad u_0 \in Q_p h^{17/8}, \quad D = \|u_0 - \Psi(p,x_0)\|_{h^{17/8}} < \delta.$$

Assume confinement in the sense of Definition 1.1. The normalized HYM flow starting at $H^{\mathrm{in}} = g_0^*(h_p e^{u_0})$ then exists in these coordinates for all $s \geq 0$. There is a selected exact trajectory (x_*, g_*) of (7), with $x_*(s) = g_*(s)x_b \in Z_p \cap B_{2r}(V)$, such that

$$H_*(s) = g_*(s)^*(h_p e^{\Psi(p,x_*(s))})$$

is an exact HYM flow on d_{x_b} and

$$\|u(s) - \Psi(p,x(s))\|_{h^{17/8}} \leq CDe^{-cs}, \tag{9}$$

$$|x(s) - x_*(s)| + d_R(g(s), g_*(s)) \leq CDe^{-cs}, \tag{10}$$

$$e^{-CDe^{-cs}}H_*(s) \ \leq H(s) \leq e^{CDe^{-cs}}H_*(s). \tag{11}$$

More precisely, with $q(s) = g(s)g_*(s)^{-1}$ and $v(s) = u(s) - \Psi(p,x(s))$, the centered relative logarithm

$$\log\Big(e^{-\Psi(p,x_*)/2}q^{*h_p}e^{\Psi(p,x)+v}qe^{-\Psi(p,x_*)/2}\Big) \tag{12}$$

has $h^{17/8}$ norm at most CDe^{-cs}.

(iii) Dependence on initial data. *For fixed p, the selected initial phase $(x_*(0), g_*(0))$ is the restriction of a C^1 map of the ambient sliced initial data (x_0, g_0, u_0), with uniform derivative bounds in the indicated Hölder and right-trivialized group norms. These bounds are uniform for small p. Nearby fixed-determinant metrics are covered by the smooth charts of Proposition 5.1.*

The constants use the fixed complementary elliptic gap and smooth local bounds. They are independent of positive finite-dimensional spectral lower bounds, ratios of positive eigenvalues of $C_x^*C_x$, and stabilizer changes. They may depend on the fixed smooth background, m, the

parameter neighborhood and the confinement radii. The chosen finite field has a complete extension agreeing with it on B_{2r} whose right-trivialized derivative bound is $L < c$. The selected lifted trajectory is unique among trajectories of this extension that track the actual lifted trajectory at some exponential rate faster than L.

The graph and its complete extension are choices. The theorem does not assert a canonical phase independent of those choices, or differentiability on a singular coarse moduli space. Its regularity statements are made before quotienting, in ambient harmonic and metric coordinates. The exponentially decaying quantity is the error transverse to the selected finite motion. The finite trajectory itself may relax slowly.

1.3. **Predecessors and proof strategy.** Finite-dimensional models already play a substantial role in HYM asymptotics. Haiden–Katzarkov–Kontsevich–Pandit [HKKP18, HKKP23] establish order comparison and algebraically determined asymptotics in their curve and quiver settings. Lam [Lam26, Theorem 8.2 and Proposition 8.6] proves locally free graded asymptotics for semistable bundles, under his stated rationality hypothesis, and bounded-distance comparison with a reduced flow. His Kuranishi construction retains the full nonabelian automorphism group and the complete obstruction [Lam26, Theorem 8.7]. These aspects of the local model are not additional assertions of the present theorem.

The comparison proved here selects an exact flow on a parabolically invariant graph, depends C^1 on the initial metric, and has relative error tending exponentially to zero. It does not supply the algebraic iterated-filtration asymptotics of the corrected finite field, and thus does not replace Lam's asymptotic classification. The two results address different parts of the dynamics.

The abstract principles are also established. Simonett's parameter-dependent invariant-manifold theorem [Sim94] provides the graph after a common parabolic realization is verified. Johnson–Latushkin–Schnaubelt [JLS14] prove exact exponentially tracking center trajectories in a parabolic setting. Differentiable stable foliations and noncompact normally hyperbolic manifolds have their own general theory [PWS13, BLZ00, BLZ08]. Our task is the geometric realization of these principles with the entire parallel group retained, including uniform estimates on the lifted phase and its reconstruction as a metric. Equivariant local bundle models and nearby-polarization HYM theory are developed by Buchdahl–Schumacher and Delloque [BS22, Del25a, Del25b].

The proof has three main steps. We first derive an exact noncommuting metric equation. Centering the derivative of the exponential removes the variable second-order coefficients, and leaves a fixed elliptic operator on the complement of all parallel Hermitian endomorphisms. Its gap persists under the allowed parameters. This gives a chosen invariant graph on the ambient harmonic space, while equivariance preserves the full integrable locus.

We next retain the group variable in the asymptotic-phase argument. Pulling the perturbed finite trajectory backward by the exact reduced flow gives a curve of finite length in $V \times G$. Slow derivative growth of the reduced flow and fast normal decay make both this curve and its initial-data derivatives converge. The limiting phase lies in G itself, so it preserves the original holomorphic orbit without inverting its possibly rank-changing differential.

Finally, the right quotient gg_*^{-1} cancels the possibly large reconstructing gauges in the relative metric. The same estimate keeps the selected coefficient trajectory inside the geometric domain. An inner-radius argument then covers nearby initial metrics. The noncommuting bicubic example in Section 6 shows how full obstruction, multiplicities, changing stabilizers and separated finite scales occur in one smooth geometric background. Singular graded

sheaves and collapsing backgrounds require additional analytic estimates; no complementary-gap assertion for them is used here.

2. Exact metric coordinates

2.1. **Moving backgrounds and the full deformation space.** We first justify the geometric identifications used in the theorem. For each fixed stable factor F_i, solve the HE equation on its determinant line with a normalized scalar potential. On trace-free Hermitian endomorphisms of F_i, the HE linearization has no kernel: a kernel element is holomorphic and hence scalar, and its pointwise trace is zero. The implicit-function theorem and elliptic regularity therefore give a smooth family $h_i(p)$ of factor HE metrics. These metrics make the factors polystable; their unchanged scalar holomorphic endomorphism algebras exclude a nontrivial polystable decomposition. They consequently remain stable. This is the moving-factor construction of [Wue26e, Proposition 5.1]. The HE existence and polystability correspondence used here are the classical ones [Don85, UY86, Kob87].

The underlying Dolbeault complex remains fixed as p varies. Its cohomology dimensions are constant, so its harmonic projections and Green operators vary smoothly. On a smaller parameter neighborhood their complementary positive spectra have a common gap. Project a basis of each central harmonic factor to the moving harmonic space and normalize by the positive square root of its Gram matrix. Tensor these identifications with the multiplicity Hom spaces. This gives the smooth G-equivariant maps ι_p used in (3), preserving the trace constraint. Every uniform assertion below is at a prescribed finite spatial and parameter regularity order on a common smaller neighborhood.

Smooth positive bundle isometries from h_p to h_0 identify the Hermitian endomorphism bundles. They commute with multiplicity automorphisms. The range $\mathfrak{a}$ of P_p is therefore fixed. Since $P_pP_0 = P_0$ and $P_0P_p = P_p$, the map

$$T_p = Q_p|_{Q_0}, \qquad T_p^{-1} = Q_0|_{Q_p}, \tag{13}$$

identifies the complementary spaces. These are smooth, K-equivariant identity-plus-finite-rank maps, bounded in every Hölder space used in the proof.

The convergent construction (3) is defined on an ambient ball, including nonintegrable points. On its zero locus $d + a_p(x)$ is integrable. We use the full construction as in [Lam26, Theorem 8.7] and [Wue26e, Lemma 2.1]. One consequence needed for large reconstructing gauges deserves an explicit proof.

Lemma 2.1. *Whenever x and gx belong to the common reconstruction ball,*

$$a_p(gx) = g\, a_p(x) g^{-1}.$$

This holds for every $g \in G$, without a bound on its norm. In particular, integrability is preserved between these endpoints.

Proof. Parallel multiplicity automorphisms commute with the Hodge operators, including nonunitary automorphisms. Expand the convergent solution of (3) into homogeneous terms $a_p(x) = \sum_{j\geq 1} a_{p,j}(x)$. Each term is obtained recursively from the equivariant bilinear operation $-d_p^* G_{2,p}(\,\cdot \wedge \cdot\,)$, and is therefore G equivariant for every g. At the two endpoints both series converge. The fixed linear operator Ad_g commutes with their sum, proving the identity. The same argument applies to κ_p; alternatively the identity conjugates the two Dolbeault operators. No intermediate point on a group path is required to lie in the small ball. □

Every sufficiently small integrable deformation has such a harmonic presentation. Impose $d_p^* a' = 0$ by a small determinant-one complex gauge orthogonal to the full parallel algebra. The linearized Coulomb equation on that complement is $-d_p^* d$, with a uniformly bounded inverse. Its right side already lies in the complement. The implicit-function theorem gives a common local slice. Integrability and Hodge decomposition then give $a' = x - d_p^* G_{2,p}(a' \wedge a')$, so contraction uniqueness identifies a' with $a_p(x)$. This construction uses a fixed elliptic complement, not the varying tangent space to a finite orbit.

2.2. **The exact sliced equation.** For small $u \in Q_p h^{19/8}(\mathrm{Herm}_0 E_0)$ put $h(u) = h_p e^u$. Even when $d_x = d + a_p(x)$ is not integrable, it determines a unique compatible unitary connection with this $(0,1)$ part. In the holomorphic presentation its $(1,0)$ part is

$$D'_{0,p} + e^{-u} D'_{0,p}(e^u) - e^{-u} a_p(x)^{*h_p} e^u. \tag{14}$$

Define the centered curvature

$$\mathcal{M}(p,x,u) = e^{u/2}\big(i\Lambda_p F_{d_x,h(u)} - \kappa_E(p) I\big) e^{-u/2}. \tag{15}$$

It is Hermitian and pointwise trace free, because the determinant connection remains the fixed HE determinant. At $x = u = 0$, $\mathcal{M} = -b_p$.

For clarity, write $\mathsf{C}_u, \mathsf{S}_u, \mathsf{B}_u$ for functions of the pointwise commutator ad_u, reserving C_x for the infinitesimal group action:

$$\mathsf{C}_u = \cosh(\mathrm{ad}_u/2), \qquad \mathsf{S}_u = \frac{\sinh(\mathrm{ad}_u/2)}{\mathrm{ad}_u/2}, \qquad \mathsf{B}_u = \mathsf{S}_u^{-1}\mathsf{C}_u, \qquad R_p(u) = P_p \mathsf{B}_u|_{\mathfrak{a}}. \tag{16}$$

The removable singularities have value one. In a unitary basis diagonalizing u, the multipliers of S_u and B_u on an off-diagonal entry are $\sinh(a)/a$ and $a\coth(a)$, where $a = (u_i - u_j)/2$. These are positive and $a\coth(a) \geq 1$. Thus $\mathsf{B}_u \geq I$ and

$$R_p(u) \geq I, \qquad \|R_p(u)^{-1}\|_{L^2(\mathfrak{a})\to L^2(\mathfrak{a})} \leq 1. \tag{17}$$

This inverse is unrelated to the spectrum of $C_x^* C_x$.

Proposition 2.2. *Define*

$$\begin{aligned} \beta(p,x,u) &= -R_p(u)^{-1} P_p \mathsf{S}_u^{-1} \mathcal{M}(p,x,u), \\ \dot{x} &= C_x \beta(p,x,u), \\ \dot{u} &= -2\mathsf{S}_u^{-1}\mathcal{M}(p,x,u) - 2\mathsf{B}_u \beta(p,x,u), \\ \dot{g} &= \beta(p,x,u) g. \end{aligned} \tag{18}$$

For classical solutions, these equations preserve $P_p u = 0$, trace and determinant. If $x(0) = g(0)x_b$ and $x_b \in Z_p$, then $x(s) = g(s)x_b$ and $x(s) \in Z_p$ as long as the coefficients stay in the reconstruction domain. The metric $H(s) = g(s)^*(h_p e^{u(s)})$ on d_{x_b} satisfies exactly

$$H^{-1}\partial_s H = -2(i\Lambda_p F_H - \kappa_E(p) I). \tag{19}$$

Proof. Projection of $\dot{u}$ onto $\mathfrak{a}$ is zero by the definition of R_p and β. The coefficient β is Hermitian and has weighted trace zero, so the last equation remains in G. Uniqueness for the linear representation equation gives $x(s) = g(s)x_b$. Lemma 2.1 gives $d_{x(s)} = g(s) d_{x_b} g(s)^{-1}$ and exact integrability.

Transport the metric velocity by g^{-1} and center by $e^{-u/2}$. Since $\dot{g}g^{-1} = \beta = \beta^{*h_p}$, the two gauge derivatives contribute $2\mathsf{C}_u\beta$. The exponential derivative contributes

$$e^{-u/2}(d\exp)_u(\dot{u})e^{-u/2} = \int_{-1/2}^{1/2} e^{t\,\mathrm{ad}_u}\dot{u}\,dt = \mathsf{S}_u\dot{u}.$$

Their sum is $-2\mathcal{M}$ by (18). Curvature transforms by the same bundle isometry, proving (19). The calculation uses neither commutativity of u and β nor commutativity of successive values of β. □

The classical calculation requires one time derivative and two spatial derivatives of u. The parabolic realization below supplies this regularity for positive times and continuity at the initial time in $h^{17/8}$. For constant $k \in K$, the transformation $(x, u, g) \mapsto (kx, kuk^{-1}, kg)$ preserves (18) and reconstructs the same metric. This is the unitary redundancy of these coordinates. Removing g instead would identify distinct parallel positive metrics at $x = 0$.

2.3. **Relative metrics and large gauges.** A right-invariant Riemannian metric on the finite-dimensional group G is complete. Its distance satisfies

$$d_R(g, g_*) = d_R(gg_*^{-1}, I).$$

The order of the two factors matters. This right quotient, rather than $g_*^{-1}g$, is what cancels the absolute gauge size in the comparison of metrics.

Lemma 2.3. *Let $\Psi : B_r(V) \to Q_p h^\tau(\mathrm{Herm}_0 E_0)$ be a C^1 map, where $\tau > 0$ is nonintegral, with $\|\Psi(x)\|_{h^\tau} \le Cr^2$ and $\|D\Psi(x)\|_{V\to h^\tau} \le Cr$. Consider metrics on the same d_{x_b},*

$$H = g^*(h_p e^{\Psi(x)+v}), \qquad H_* = g_*^*(h_p e^{\Psi(x_*)}), \qquad x = gx_b, \quad x_* = g_* x_b,$$

with $x, x_* \in B_r$ and $\|v\|_{h^\tau}$ and $d_R(g, g_*)$ sufficiently small. The centered logarithm

$$\log\left(e^{-\Psi(x_*)/2} q^{*h_p} e^{\Psi(x)+v} q e^{-\Psi(x_*)/2}\right), \qquad q = gg_*^{-1},$$

has h^τ norm at most $C(\|v\|_{h^\tau} + d_R(g, g_*))$. In particular, $e^{-\eta}H_* \le H \le e^\eta H_*$ for this same bound η. The constants are independent of the absolute gauge norms and of the stabilizers.

Proof. In an identity chart, $\|q - I\| \le C d_R(g, g_*)$. Since $x = qx_*$, it follows that $|x - x_*| \le Cr\, d_R(g, g_*)$, without inverting C_x. The segment between x and x_* lies in the ambient ball, so the graph derivative bound gives $\|\Psi(x) - \Psi(x_*)\|_{h^\tau} \le Cr^2 d_R(g, g_*)$. The matrix q is parallel for the split connection, with uniformly bounded spatial norms in the identity chart. Multiplication, exponential and logarithm are smooth operations in the little-Hölder algebra. The displayed logarithm vanishes at $q = I, v = 0$, which gives its norm estimate. Its C^0 operator bound implies the metric-order statement after pullback by g_*. No estimate on g or g_* enters. □

3. The complementary equation and its invariant graph

3.1. **The real elliptic operator.** Write $D_{0,p} = D'_p + d$ for the split Chern connection. On trace-free Hermitian zero-forms set

$$L_p = \tfrac{1}{2}(\Delta_{D'_p} + \Delta_d). \tag{20}$$

Its real quadratic form is half the sum of the two derivative energies. Its kernel therefore consists of parallel Hermitian endomorphisms and equals $\mathfrak{a}$. Indeed, a parallel endomorphism is holomorphic, the cross-Hom spaces between the distinct stable factors vanish, and each stable factor has only scalar holomorphic endomorphisms. Conversely, all multiplicity endomorphisms are parallel. The restriction of L_p to Q_p has a common positive gap for small p.

Lemma 3.1. *On the ambient coefficient ball the exact expression*

$$\mathsf{S}_u^{-1}\mathcal{M}(p,x,u) = L_p u - b_p + \mathcal{G}_p(x,u,\nabla u) \tag{21}$$

has a smooth remainder depending only on the first spatial jet of u. For every p, $\mathcal{G}_p(0,0) = 0$ and $D_{(x,u)}\mathcal{G}_p(0,0) = 0$.

Proof. In (14), second spatial derivatives occur only when $e^{-u}D'_p(e^u)$ is differentiated. If W denotes the contracted second jet, its centered coefficient is $e^{-u/2}(d\exp)_u(W)e^{-u/2} = \mathsf{S}_u W$. Multiplying by S_u^{-1} cancels this coefficient exactly. Every remaining derivative of u has order at most one. The finite form $a_p(x)$ and its spatial derivatives are smooth and uniformly controlled on the chosen coefficient ball.

The constant term is $-b_p$. Put $K_p = i\Lambda_p F_{D_{0,p}}$. Direct differentiation and the zero-form Kähler identity give

$$\begin{aligned} D_u\mathcal{M}(p,0,0)u &= i\Lambda_p\bar\partial D'_p u + \tfrac{1}{2}[u,K_p] \\ &= \Delta_{D'_p}u - \tfrac{1}{2}[K_p,u] = \tfrac{1}{2}(\Delta_{D'_p} + \Delta_d)u. \end{aligned} \tag{22}$$

Here $\Delta_d = \Delta_{D'_p} - [K_p,\cdot]$. The commutator is needed when the factor slopes differ. Since $\mathsf{S}_u^{-1} = I + O(u^2)$, it does not alter the linearization, including in the presence of the constant b_p. The coefficient derivative vanishes: contraction of $D_{0,p}(x-x^*)$ is a combination of $d_p^* x$ and its adjoint, which vanish for harmonic x. Subtracting the constant and linear parts proves the statement. □

Substitution in (18) yields

$$\begin{aligned} \beta(p,x,u) &= R_p(u)^{-1}(b_p - P_p\mathcal{G}_p), \\ \dot u &= -2L_p u + \mathcal{N}_p(x,u), \\ \mathcal{N}_p &= -2\mathcal{G}_p + 2(I - \mathsf{B}_u R_p(u)^{-1})b_p + 2\mathsf{B}_u R_p(u)^{-1}P_p\mathcal{G}_p. \end{aligned} \tag{23}$$

In the middle term I includes $\mathfrak{a}$ in all sections. The projection identity in Proposition 2.2 shows that $\mathcal{N}_p$ takes values in Q_p. Both $\mathsf{B}_u - I$ and $R_p(u) - I$ are quadratic, so

$$D\mathcal{N}_p(0,0) = 0, \qquad D_u\beta(p,0,0) = 0. \tag{24}$$

The finite linearization is $\Lambda_p x = C_x b_p$, with norm tending to zero as $p \to 0$.

3.2. **A common parabolic domain.** Use (13) to work on the fixed pair

$$\mathcal{X}_0 = V_{\mathbb{R}} \oplus Q_0 h^{3/8}, \qquad \mathcal{X}_1 = V_{\mathbb{R}} \oplus Q_0 h^{19/8}.$$

The continuous interpolation spaces needed below are

$$\begin{aligned} \mathcal{X}_\beta &= (\mathcal{X}_0,\mathcal{X}_1)^0_{5/8,\infty} = V_{\mathbb{R}} \oplus Q_0 h^{13/8}, \\ \mathcal{X}_\alpha &= (\mathcal{X}_0,\mathcal{X}_1)^0_{7/8,\infty} = V_{\mathbb{R}} \oplus Q_0 h^{17/8}. \end{aligned} \tag{25}$$

We use the interpolation convention of [SS14, Proposition 2.2(c)]. One derivative of $h^{13/8}$ belongs to the multiplication algebra $h^{5/8}$, which embeds in $h^{3/8}$. Smooth multiplication and composition therefore make the first-jet nonlinearities smooth $\mathcal{X}_\beta \to \mathcal{X}_0$ maps. No restriction on $\dim_{\mathbb{R}} X$ is required for this Hölder algebra estimate. Smooth finite-rank projections preserve these interpolation spaces. The exponents fix a convenient interpolation scale; no sharp regularity threshold is asserted.

The transported operator $\operatorname{diag}(0, 2L_p|_{Q_p})$ on $\mathcal{X}_1 \subset \mathcal{X}_0$ has continuous maximal regularity. We spell out the bundle realization of the result used here. Complexify the real bundle and

choose a smooth complementary bundle making their sum trivial. Extend $2L_p$ by a connection Laplacian on that complement with the same scalar principal symbol. The resulting normally elliptic system has domain $h^{19/8}$ on $h^{3/8}$ by the local resolvent construction and Banach-valued extension of [SS14, Remark 3.3, Theorem 3.6 and Proposition 3.7]. The original summand is reducing. Projection away from the parallel kernel commutes with the operator, so its linear solutions restrict to the required complement. A finite-dimensional zero block also has maximal regularity. Uniform ellipticity and smooth dependence on a compactly contained parameter neighborhood give common constants.

In particular, on $E_{0,p} = Q_p h^{3/8}$ and $E_{\alpha,p} = Q_p h^{17/8}$, the semigroup of $A_p = 2L_p|_{Q_p}$ satisfies

$$\|e^{-sA_p}\|_{E_\alpha \to E_\alpha} \le Me^{-\sigma s}, \qquad \|e^{-sA_p}\|_{E_0 \to E_\alpha} \le M(1 + s^{-7/8})e^{-\sigma s}. \tag{26}$$

Elliptic regularity identifies its spectrum with the smooth L^2 spectrum. A common resolvent contour gives the exponential bound from the fixed complementary gap; interpolation gives the smoothing factor. The restricted semigroup is strongly continuous on the little-Hölder initial space. The parameter-dependent second-order term has remained in L_p throughout this argument.

3.3. **The invariant graph.**

Proposition 3.2. *For each prescribed finite $m \ge 6$, after shrinking the parameter and coefficient neighborhoods there is a jointly C^m, K-equivariant map $\Psi(p,x) \in Q_p h^{19/8}$ whose graph is locally invariant for (18), and*

$$\Psi(p,0) = D_x\Psi(p,0) = 0, \qquad \|\Psi(p,x)\|_{h^{19/8}} \le C|x|^2, \qquad \|D_x\Psi(p,x)\|_{V \to h^{19/8}} \le C|x|. \tag{27}$$

Its exact coefficient field is (7), with $\beta_0(p,x) = \beta(p,x,\Psi(p,x))$. Every integrable trajectory, lifted by $\dot g = \beta_0(p,x)g$, reconstructs an exact HYM solution.

Proof. At $p = 0$, the finite space $V_{\mathbb{R}}$ is the entire center space of the (x,u) equation. The complementary linearization $-2L_0|_{Q_0}$ is strictly stable. The equations (23)–(26) verify the common domain, smoothness, interpolation and continuous maximal-regularity hypotheses of [Sim94, Theorem 2.1]. That parameter-dependent theorem gives a chosen C^m graph with values in the stable domain. The equilibrium $(x,u) = (0,0)$ exists for each parameter, and its construction gives $\Psi(p,0) = 0$.

Tangency for every nearby parameter requires an additional argument. Set $T = D_x\Psi(p,0)$. Differentiating the exact invariance equation, using (24), gives

$$T\Lambda_p = -2L_pT.$$

The finite operator Λ_p is diagonalizable with real eigenvalues equal to differences of central block constants in b_p. They tend to zero. Apply the displayed identity to each eigenvector; the disjoint finite and normal spectra force $T = 0$. Taylor estimates on a smaller parameter product give (27).

Choose invariant norms and a K-invariant coefficient cutoff in the fixed-point construction. Acting by $k \in K$ gives another fixed point in the same uniqueness class, hence the same graph. This proves equivariance without averaging invariant graphs. Substitution gives the exact coefficient field, and Proposition 2.2 gives reconstruction. □

The graph depends on the construction and on the chosen finite regularity order. We use only its stated C^m regularity. Its invariance equation is

$$D_x\Psi(p,x)C_x\beta_0(p,x) = -2L_p\Psi(p,x) + \mathcal{N}_p(x,\Psi(p,x)). \tag{28}$$

The left side distinguishes it from a solution of a stationary complementary curvature equation.

3.4. **The leading moment field.**

Proposition 3.3. *The graph field satisfies* (8)*, with the moment convention* (5)*.*

Proof. Transfer the connection of $h_p e^{\Psi}$ isometrically to h_p. Its $(0,1)$ perturbation has the expansion

$$\widetilde{a} = x + a_2(x) - \tfrac{1}{2} d\Psi_2(p,x) + O(|x|^3), \qquad a_2(x) = -d_p^* G_{2,p}(x \wedge x).$$

The quadratic correction is orthogonal to every harmonic one-form: its two terms are coexact and exact, respectively. For a parallel Hermitian endomorphism ξ, integration of curvature gives the exact identity

$$\langle P_p \mathcal{M}(p,x,\Psi), \xi \rangle = -\langle b_p, \xi \rangle + \mathrm{Re}\langle [\xi, \widetilde{a}], \widetilde{a} \rangle. \tag{29}$$

The derivative term integrates to zero because ξ is parallel. In a unitary coframe the remaining term is a sum of $[A_j, A_j^*]$, and $\mathrm{tr}(\xi[A_j, A_j^*]) = \mathrm{Re}\,\mathrm{tr}([\xi, A_j]A_j^*)$. This also verifies the sign and requires no integrability assumption on the ambient coefficient.

The action $[\xi, \cdot]$ preserves harmonic forms and is self-adjoint for the real one-form inner product. Consequently both cubic cross terms between x and the quadratic correction vanish. Equation (29) becomes $P_p \mathcal{M} = -b_p + \mu_p^\sharp + O(|x|^4)$. Moreover $\mathsf{S}_\Psi^{-1} - I$ and $R_p(\Psi)^{-1} - I$ are $O(|x|^4)$, whereas $\mathcal{M}(p,x,\Psi) = O(|p| + |x|^2)$ in the required spatial norm. The formula for β in (18) therefore retains the quartic remainder. The vanished jets and the C^m bounds give its first two differentiated estimates.

Finally, $d\langle \mu_p^\sharp, \xi \rangle[v] = 2\,\mathrm{Re}\langle C_x \xi, v \rangle$. Differentiating $\frac{1}{4}\|\mu_p^\sharp - b_p\|^2$ gives the leading field in Theorem 1.2. □

The factor-rank and volume weights are part of the Riesz identification in (5). Changing the moment normalization changes the scalar time factor. In all tracking statements we use the exact field β_0, including its remainder.

4. Differentiable phase in the reconstruction group

The invariant graph controls the distance to the graph above the evolving coefficient. To select one exact trajectory on it, we retain the reconstruction group. The following phase argument is formulated for a finite representation coupled to an exponentially stable equation. It is a concrete version of the asymptotic-phase mechanism; its group-valued estimates are what will be used for relative HYM metrics.

4.1. **The finite extension and its derivatives.**

Let V be a finite-dimensional real inner-product space carrying a representation of a closed matrix group G. Give G a complete right-invariant Riemannian metric. A tangent variation δg is measured by $\xi = \delta g\, g^{-1}$ in a fixed Lie algebra norm. Let $\beta : V \to \mathfrak{g}$ be bounded and C^2, and put $f(x) = C_x \beta(x)$. Suppose f is globally Lipschitz and the right-trivialized linearization

$$\mathcal{A}(x)(w,\xi) = (Df(x)w, D\beta(x)w + [\beta(x), \xi]) \tag{30}$$

satisfies $\|\mathcal{A}(x)\| \le L$ and $\|D_x \mathcal{A}(x)\| \le K_0$. The small quantity needed below is the upper bound L, not a positive lower spectral bound.

The complete finite flow has the form

$$\Phi_s(x,g) = (\phi_s(x), B_s(x) g).$$

Global Lipschitz growth controls x, and bounded β controls both the matrix and its inverse on every finite time interval. For a variation, $w = \delta x$ and $\xi = \delta g\, g^{-1}$ satisfy

$$\dot{w} = Df(x)w, \qquad \dot{\xi} = D\beta(x)w + [\beta(x), \xi].$$

Thus the right-trivialized derivative $T_s(x)$ depends on x but not on g. The fundamental matrix equation, and then its x derivative, give for either sign of s

$$\|T_s(x)\| \le e^{L|s|}, \qquad \|D_x T_s(x)\| \le K_0|s|e^{2L|s|}. \tag{31}$$

For the second inequality, variation of constants uses $\|D\phi_s\| \le e^{L|s|}$; for negative times apply the same argument to the reversed equation. Integration along paths gives the corresponding Lipschitz bound for Φ_s in the product distance on $V \times G$.

4.2. **Statement of the phase estimate.** Let $E_1 \subset E_\alpha \subset E_0$ be continuously embedded Banach spaces, $0 < \alpha < 1$. Suppose $-A$ generates a C_0 semigroup on E_0 whose restriction is also C_0 on E_α, and

$$\|e^{-sA}\|_{E_\alpha \to E_\alpha} \le Me^{-\sigma s}, \qquad \|e^{-sA}\|_{E_0 \to E_\alpha} \le M(1 + s^{-\alpha})e^{-\sigma s}. \tag{32}$$

On $V \times B_{E_\alpha}(\rho)$ let $h = (h_x, h_g)$ and $\mathcal{F}$ be C^2, with values in $V \oplus \mathfrak{g}$ and E_0, respectively, and uniform bounds through order two. Assume

$$\begin{gathered} h(x,0) = 0, \quad \mathcal{F}(x,0) = 0, \quad \|D_v h\| + \|D_v \mathcal{F}\| \le \epsilon, \\ \|D_x h(x,v)\| + \|D_x \mathcal{F}(x,v)\| \le K_0 \|v\|_{E_\alpha}. \end{gathered} \tag{33}$$

Consider

$$\dot{x} = f(x) + h_x(x,v), \qquad \dot{g} = (\beta(x) + h_g(x,v))g, \qquad \dot{v} = -Av + \mathcal{F}(x,v). \tag{34}$$

These bounds are in fixed finite and Banach operator norms; none is a bound in the absolute matrix coordinates of g.

Proposition 4.1. *Fix the constants in (32) and the uniform C^2 bounds. If L, ϵ are sufficiently small, there is $\delta > 0$ such that for $D = \|v_0\|_{E_\alpha} < \delta$, (34) has a global mild solution and a phase*

$$\Pi(x_0, g_0, v_0) = \lim_{T \to \infty} \Phi_{-T}(x(T), g(T)) \quad \text{in } V \times G. \tag{35}$$

The map Π is C^1 with uniform derivative bounds in right-trivialized norms. For $a = \sigma/2$ and $Y_*(s) = \Phi_s \Pi$,

$$\|v(s)\|_{E_\alpha} \le CDe^{-as}, \qquad d((x(s), g(s)), Y_*(s)) \le C\epsilon De^{-as}. \tag{36}$$

The selected trajectory is unique among complete finite trajectories tracking at $O(e^{-qs})$ for some $q > L$. If $v_0 = 0$, its phase is (x_0, g_0). The constants are uniform for families with the same assumptions, independent of g_0 and of finite spectral gaps.

If $h_x = C_x h_g$ and $x_0 = g_0 x_b$, then $x_*(s) = g_*(s)x_b$.

Proof. Normal decay and continuation. The weighted convolution constant

$$J_a = M \int_0^\infty (1 + t^{-\alpha})e^{-(\sigma - a)t}\, dt$$

is finite. Duhamel's formula implies

$$\sup_{s \le T} e^{as}\|v(s)\| \le MD + \epsilon J_a \sup_{s \le T} e^{as}\|v(s)\|.$$

Choose $\epsilon J_a \le 1/2$ and $2M\delta < \rho/2$. Then $\|v(s)\| \le 2MDe^{-as}$, so the solution never reaches the normal boundary. The globally Lipschitz finite field and bounded perturbations exclude

finite-time escape of x or g. Local mild existence and continuation follow by contraction with the integrable smoothing kernel $t^{-\alpha}$. The same argument gives local C^1 dependence in the stated initial topology.

Convergence in the group and tracking. Write $Y(s) = (x(s), g(s))$ and $p_T = \Phi_{-T} Y(T)$. Its velocity in the invariant frames at p_T is

$$\dot{p}_T = T_{-T}(x(T)) h(x(T), v(T)) =: w_T. \tag{37}$$

By (31), $\|w_T\| \le 2M\epsilon D e^{-(a-L)T}$. For $L < a$, the curve p_T has finite length, so completeness of $V \times G$ gives (35). Its distance from (x_0, g_0) is at most $2M\epsilon D/(a-L)$.

For fixed s, differentiate $\Phi_{s-T} Y(T)$ for $T \ge s$. The velocity bound is

$$2M\epsilon D e^{L(T-s)} e^{-aT}.$$

Integration gives

$$d(Y(s), \Phi_s \Pi) \le \frac{2M\epsilon D}{a-L} e^{-as}.$$

If two finite trajectories have distance $O(e^{-qs})$ with $q > L$, applying Φ_{-s} forces their initial distance to be zero. A second trajectory tracking the actual solution at such a rate differs from the selected one at rate $\min(a, q) > L$, proving the stated uniqueness. Existence at every rate $q > L$ is not asserted.

Differentiability of the limit. The (x, v) system is independent of g. Set $\gamma = \sigma/16$ and require $L < \gamma/4$. For variations of its initial data write $W = \delta x$ and $Z = \delta v$. Then

$$\dot{W} = (Df + D_x h_x) W + D_v h_x Z$$

and

$$Z(s) = e^{-sA} Z_0 + \int_0^s e^{-(s-t)A} (D_x \mathcal{F}\, W + D_v \mathcal{F}\, Z)(t)\, dt.$$

Define $U_T = \sup_{s \le T} e^{-\gamma s} |W(s)|$ and $V_T = \sup_{s \le T} e^{(a-\gamma)s} \|Z(s)\|$. The normal convolution estimate, with ϵ smaller if necessary, gives

$$V_T \le C\|Z_0\| + CK_0 D U_T.$$

Variation of constants for W uses $\int_0^\infty \|v(t)\| dt \le 2MD/a$, so $D_x h_x$ has integrable norm. It gives $U_T \le C|W_0| + C\epsilon V_T$. Choose δ so that the resulting $C\epsilon K_0 D U_T$ term is absorbed. Uniformly in T,

$$\begin{aligned} |W(s)| &\le C e^{\gamma s} (|W_0| + \epsilon \|Z_0\|), \\ \|Z(s)\| &\le C e^{-(a-\gamma)s} (\|Z_0\| + D|W_0|). \end{aligned} \tag{38}$$

The convolution is finite because $a - \gamma < \sigma$.

Differentiate the right-trivialized velocity (37) with respect to (x_0, v_0):

$$\delta w_T = (D_x T_{-T})(x(T))[W(T)] h(x(T), v(T)) + T_{-T}(x(T))(D_x h\, W + D_v h\, Z). \tag{39}$$

Its norm is bounded by a constant times

$$\left[\epsilon D(1+T) e^{-(a-2L-\gamma)T} + D e^{-(a-L-\gamma)T} + \epsilon e^{-(a-L-\gamma)T}\right] \cdot (|W_0| + \|Z_0\|). \tag{40}$$

This bound is integrable, uniformly on a smaller initial normal ball.

To pass from this estimate to derivatives of p_T, use its Euclidean component and an identity chart for its group component after right translation by g_0^{-1}. The phase-length estimate keeps $q_T = p_T^G g_0^{-1}$ inside that chart for small δ. It satisfies $\dot{q}_T = w_T^G q_T$ and

$$\partial_T(\delta q_T q_T^{-1}) = [w_T^G, \delta q_T q_T^{-1}] + \delta w_T^G.$$

Integrability of w_T and δw_T implies uniform convergence of these derivatives. The limits are continuous by dominated convergence and finite-time continuous dependence. Thus the limiting phase is C^1 in (x_0, v_0). For arbitrary g_0, right equivariance gives

$$\Pi^G(x_0, g_0, v_0) = \Pi^G(x_0, I, v_0)g_0.$$

Its derivative in g_0, in right-trivialized norms, is conjugation by an element in the same fixed identity chart, and is uniformly bounded.

Preservation of the orbit. When $h_x = C_x h_g$, both perturbed and unperturbed equations preserve $x = gx_b$. Every finite-time phase point satisfies $p_T^V = p_T^G x_b$. Since the limit converges in the group itself, this equality passes to the limit and then along the finite flow. No inverse of C_x occurs. □

5. Geometric tracking and initial metrics

5.1. **Verification of the phase hypotheses.** Choose the graph of Proposition 3.2 on $B_{4r}(V)$. Set $v = u - \Psi(p, x)$ and define

$$\begin{aligned} h_g(p, x, v) &= \beta(p, x, \Psi(p, x) + v) - \beta_0(p, x), \\ h_x(p, x, v) &= C_x h_g(p, x, v), \\ \mathcal{F}_p(x, v) &= \mathcal{N}_p(x, \Psi + v) - \mathcal{N}_p(x, \Psi) - D_x\Psi(p, x)C_x h_g(p, x, v). \end{aligned} \tag{41}$$

The last subtraction follows from the invariance equation (28). In these variables the exact normal equation is $\dot{v} = -A_p v + \mathcal{F}_p(x, v)$. Its remainder contains no second spatial derivative of v. All three functions in (41) vanish at $v = 0$. Smoothness and (24) imply, on smaller coefficient and normal balls,

$$\|D_v h\| + \|D_v \mathcal{F}_p\| \le \epsilon, \qquad \|D_x h\| + \|D_x \mathcal{F}_p\| \le K_r \|v\|_{E_\alpha}. \tag{42}$$

Normal remainders are measured from E_α to E_0. Uniform bounds through order two follow from Hölder multiplication and composition estimates on these balls; compactness of an infinite-dimensional normal ball is not used.

For phase selection we need a complete finite extension. Choose a K-invariant smooth coefficient cutoff χ_r equal to one on B_{2r} and supported in B_{3r}, and put

$$\widehat{\beta}_0 = b_p + \chi_r(\beta_0 - b_p), \qquad \widehat{h}_g = \chi_r h_g, \qquad \widehat{h}_x = C_x \widehat{h}_g, \qquad \widehat{\mathcal{F}}_p = \chi_r \mathcal{F}_p. \tag{43}$$

The supported differences are extended by zero. The central drift outside the support is retained. To see the relevant bounds explicitly, let $B(x) = \beta_0(p, x) - b_p$. Proposition 3.3 gives, on the support,

$$|B| \le Cr^2, \qquad |DB| \le Cr, \qquad |D^2 B| \le C.$$

Since $|D^j \chi_r| \le C_j r^{-j}$, the same bounds hold for $\widehat{\beta}_0 - b_p$ through order two. For $\widehat{f}(x) = C_x \widehat{\beta}_0(x)$, $|D\widehat{f}| \le C(|p| + r^2)$. The full right-trivialized group derivative also contains $D\widehat{\beta}_0$, and hence admits the bound

$$L \le C(|p| + r). \tag{44}$$

The bracket term in (30) obeys the same bound. Outside the support the finite field is the complete central linear drift, and the group coefficient is bounded everywhere.

Multiplication by χ_r preserves the small normal derivative bounds in (42). Coefficient derivatives of the cutoff introduce powers of r^{-1}, but their factors vanish at $v = 0$, so the first coefficient derivatives remain bounded by $K_r\|v\|$. Second derivative constants are finite for the chosen radius. Proposition 4.1 does not require these constants to remain bounded as $r \to 0$. There is no dependence on a finite orbit rank or spectral ratio. We can therefore choose the parameter and coefficient radii to make L small relative to σ, then the normal radius to make ϵ small, and finally the initial displacement threshold δ.

5.2. **Proof of the main theorem.**

Proof of Theorem 1.2. Propositions 3.2 and 3.3 give part (i). Equations (26), (41) and (43) verify Proposition 4.1 with $\alpha = 7/8$, uniformly in p. It supplies a global solution of the extended system, normal exponential decay and a group-valued phase.

Before any exit from the geometric domain, uniqueness identifies this solution with the sliced HYM system. Confinement prevents coefficient exit. The quadratic graph bound and $\|v(s)\|_{h^{17/8}} \le CDe^{-cs}$ prevent normal exit after reducing r and δ. The group coefficient is bounded, so g and g^{-1} cannot escape in finite time. The mild continuation argument with the integrable smoothing kernel excludes a remaining finite-time exit in the initial Hölder space. Thus the geometric coordinate interval is infinite.

The phase estimate gives $|x(s)-x_*(s)| \le CDe^{-cs}$. Require $C\delta < r/2$. Since $|x(s)| \le r$, the selected trajectory remains inside B_{2r}, where the finite extension agrees with the exact graph field. The orbit-preserving conclusion of Proposition 4.1 gives $x_*(s) = g_*(s)x_b$. Lemma 2.1 then preserves the entire integrability equation for both trajectories. Proposition 2.2 reconstructs actual HYM flows on the same d_{x_b}.

For initial data in $h^{17/8}$, the common parabolic realization gives classical equations at positive times and continuity in that space at time zero. The reconstructed metric equation and its initial value follow by this continuation. Smooth initial metrics are included. Lemma 2.3, applied on the larger coefficient ball, gives (12) and (11). The remaining estimates are those of Proposition 4.1. Its C^1 statement combines with the smooth change $v_0 = u_0 - \Psi(p, x_0)$ to prove part (iii). Its uniqueness class and uniformity give the final assertions. Proposition 5.1 below supplies the stated interpretation in initial metric charts. □

5.3. **Coverage of nearby metrics.**

Proposition 5.1. *There is $r_{\mathrm{in}} \in (0, r/2)$ such that for $|x_b| < r_{\mathrm{in}}$ every fixed-determinant metric sufficiently close to h_p in $h^{17/8}$ has a unique local representation*

$$H^{\mathrm{in}} = (e^{a_0})^*(h_p e^{u_0}), \qquad a_0 \in \mathfrak{a}, \quad u_0 \in Q_p h^{17/8}, \tag{45}$$

with a_0, u_0 small, $x_0 = e^{a_0}x_b \in B_r$, and $\|u_0 - \Psi(p, x_0)\|_{h^{17/8}} < \delta$. The representation and its inverse are smooth with uniform local derivative bounds in p, independent of the stabilizer of x_b.

The same conclusion holds in a relative chart about a possibly large gauge $\bar{g}$ provided $|\bar{g}x_b| < r_{\mathrm{in}}$. The chart uses the metric pulled back by $\bar{g}^{-1}$ and has constants independent of the absolute norm of $\bar{g}$.

Proof. In an h_p-unitary presentation consider

$$\mathcal{J}_p(a, u) = \log(e^a e^u e^a) : \mathfrak{a} \oplus Q_p h^{17/8} \longrightarrow h^{17/8}(\mathrm{Herm}_0 E_0).$$

The matrix product is positive Hermitian with determinant one. Smooth composition in the Hölder algebra makes $\mathcal{J}_p$ smooth. Its derivative at zero is $(a,u)\mapsto 2a+u$, with uniformly bounded inverse $z\mapsto(\frac{1}{2}P_pz,Q_pz)$. The parameter-dependent inverse-function theorem gives a common metric neighborhood.

The representation action near the identity sends $B_{r_{\mathrm{in}}}$ into $B_{2r_{\mathrm{in}}}$. Choose r_{in} after the tracking threshold so that $C(2r_{\mathrm{in}})^2<\delta/2$ in the graph bound. Reduce the metric neighborhood to ensure $\|u_0\|<\delta/2$. The triangle inequality gives the required initial normal displacement. For a large gauge, apply the same factorization to the pulled-back metric at coefficient $\bar{g}x_b$, and return to the original bundle with $g_0=e^{a_0}\bar{g}$. The estimates are in the relative frame and introduce no factor $\|\bar{g}\|$. □

For integrable x_b, restriction of the ambient phase map to these metric charts is C^1. Confinement remains the hypothesis needed for its all-time geometric interpretation.

5.4. **Special cases and scope.** For diagonal multiplicity groups, the slice equations reduce to those used in the two-step metric response and four-line phase constructions [Wue26a, Section 6], [Wue26b, Sections 3–5]. In a positive diagonal frame $g=e^Y$, right group differences become differences of Y. Theorem 1.2 therefore recovers their basic confined phase and relative metric estimates when the same graph construction is chosen. Their sharper size factors and differentiated refinements remain separate estimates.

The confined wall and triangular unstable trajectories of [Wue26c, Sections 3–5] also fit the present regime after the stated radius reductions. Their gauges may diverge, while their harmonic coefficients stay small. The relative reconstruction lemma explains why this does not obstruct metric tracking in the general noncommuting setting. Acyclic-support coercivity, global entry and confinement, and precise logarithmic or linear drift require the additional arguments of those special families.

The stationary balancing and harmonic quotient comparisons of [Wue26d, Wue26e] do not supply a parabolically invariant graph. The present proof uses the full Kuranishi construction and moving stable-factor geometry, but constructs its dynamical graph and phase independently of those stationary comparisons. No assertion about an evolving moduli-space metric follows merely from the relative metric estimate proved here.

6. An obstructed repeated-factor application

6.1. **The geometric input.** We use the stable bundle constructed in [Wue26e, Section 6]. We recall its geometry and deformation equations to specify the entire background of the application. On a base $\mathbb{P}^2$, choose smooth cubic curves defined by R,P,Q, meeting pairwise transversely and with no common point. Fix $p_0\in\{P=Q=0\}$. In $\mathbb{P}^2_{[x:y:z]}\times\mathbb{P}^2$ set

$$X=\{x^3R+y^3P+z^3Q=0\},\qquad C=\{x=0\}\times\{p_0\}. \tag{46}$$

The threefold is smooth: if all fiber coordinates are nonzero, the fiber derivatives would force $R=P=Q=0$; with only one or two nonzero coordinates, smoothness or transversality of the corresponding base cubics prevents the total differential from vanishing. Adjunction and the restriction sequence give $K_X\simeq\mathcal{O}_X$ and $H^1(X,\mathcal{O}_X)=H^2(X,\mathcal{O}_X)=0$.

The Serre construction [Arr07], used for obstructed bundles by Thomas [Tho99], gives a locally free rank-two bundle

$$0\longrightarrow\mathcal{O}_X\longrightarrow E\longrightarrow I_C\mathcal{O}_X(-2,2)\longrightarrow 0,\qquad A=E(1,-1),\qquad \det A=\mathcal{O}_X. \tag{47}$$

Let ω_1, ω_2 denote the restricted hyperplane classes. The geometric input supplies stability of A for $N\omega_1 + \omega_2$, $1 < N < 1 + \sqrt{3}$, and the full fixed-determinant analytic deformation ring

$$\mathbb{C}\{a, b\}/(a^2 b, ab^2). \tag{48}$$

In particular, $\dim \operatorname{Ext}^1(A, A) = 2$, the quadratic obstruction vanishes, and the cubic obstruction is essential.

The calculations behind these assertions are concrete. With $U = Q/R$ and $V = P/R$ near p_0, the local equation is $x^3 + Vy^3 + Uz^3 = 0$. The full ambient line chart $x = ay + bz$, $U = c$, $V = d$ gives coefficients

$$a^3 + d, \qquad 3a^2 b, \qquad 3ab^2, \qquad b^3 + c.$$

Eliminating c, d gives (48). The normal bundle is $\mathcal{O}_C(1) \oplus \mathcal{O}_C(-3)$. The local Serre generator is nowhere zero, and $\mathcal{O}_X(2, -2)$ is acyclic, proving local freeness and the required global extension. The vanishings $h^0(E) = 1$ and $H^1(E) = 0$ identify the curve and bundle deformation functors, including their nonreduced analytic structure: the section lifts uniquely up to a unit, and its regular zero scheme is the inverse construction. For stability, the twists of E with sections lie in the three upper quadrants based at $(0, 0), (2, -1), (3, -2)$. Their first-Chern degrees against $(N\omega_1 + \omega_2)^2$ are respectively $6(N^2 - 1)$, $6(2N + 1)$ and $6(2 + 2N - N^2)$, all positive on the stated interval. These are the established geometric inputs; the dynamical application below uses the full deformation problem.

Fix a Kähler form in $2\omega_1 + \omega_2$ and the normalized HE metric of A, and put $E_0 = A \otimes \mathbb{C}^3$. Then

$$G = \mathrm{SL}(3, \mathbb{C}), \quad K = \mathrm{SU}(3), \quad \dim_{\mathbb{R}} \mathfrak{a} = 8, \qquad V = \operatorname{Ext}^1(A, A) \otimes \operatorname{End} \mathbb{C}^3, \quad \dim_{\mathbb{C}} V = 18. \tag{49}$$

The determinant condition removes no degree-one harmonic direction because $H^1(X, \mathcal{O}_X) = 0$. There is one distinct stable factor, so $b_p = 0$ for every nearby polarization.

Corollary 6.1. *Theorem 1.2 applies to every integrable point of the full eighteen-coordinate deformation problem at $A \otimes \mathbb{C}^3$, with common coefficient and normal radii and its stated confinement hypothesis. Its phase and relative tracking constants remain uniform through all stabilizer changes and arbitrarily separated positive finite scales.*

Proof. The construction verifies a fixed smooth compact Kähler background, local freeness, stability of the factor and the determinant normalization. All multiplicity directions in (49) are retained. Theorem 1.2 imposes no restriction on the order of the obstruction or the rank of C_x, so applies to the entire zero locus of the actual canonical κ_p. □

The matrix obstruction in this statement is defined by (3). It is not obtained by substituting matrices into the scalar equations (48). On the scalar subspace $x = u \otimes I_3$, the canonical recursion retains the cubic obstruction. A quadratic truncation would therefore admit false integrable points.

6.2. **Noncommuting integrable coefficients.** The Calabi–Yau duality supplies a useful additional fact for this two-dimensional deformation space.

Lemma 6.2. *Every product of two elements of $\operatorname{Ext}^1(A, A)$ vanishes in $\operatorname{Ext}^2(A, A)$.*

Proof. Take a basis a, b of $\operatorname{Ext}^1(A, A)$. The zero quadratic obstruction gives $a^2 = b^2 = 0$ and $ab + ba = 0$. Calabi–Yau trivialization and the Serre trace give a perfect pairing

$$\operatorname{Ext}^2(A, A) \times \operatorname{Ext}^1(A, A) \longrightarrow \mathbb{C}, \qquad (u, v) \longmapsto \operatorname{Tr}(uv).$$

Graded cyclicity gives $\mathrm{Tr}(aba) = \mathrm{Tr}(aab) = 0$: moving the last degree-one factor past the first two has sign $(-1)^{1\cdot 2} = +1$. Also $\mathrm{Tr}(abb) = 0$. Thus ab pairs to zero with the basis a, b, and perfection implies $ab = 0$. Then $ba = 0$, and bilinearity proves the result. □

For harmonic representatives α, β, the lemma implies $H_2(\alpha \wedge \beta) = H_2(\beta \wedge \alpha) = 0$, as well as the square identities. The trace in the proof is the Serre trace, or integration of the bundle trace against the holomorphic volume form. Zero scalar quadratic obstruction alone would not imply this product vanishing in an arbitrary deformation problem.

Choose α, β orthonormal in harmonic L^2. For small complex parameters s, t, set

$$x_{s,t} = s\alpha \otimes E_{12} + t\beta \otimes E_{23}, \qquad B = d^* G_2(\alpha \wedge \beta). \tag{50}$$

Proposition 6.3. *The exact canonical reconstruction is*

$$a(x_{s,t}) = s\alpha \otimes E_{12} + t\beta \otimes E_{23} - stB \otimes E_{13}, \qquad \kappa(x_{s,t}) = 0. \tag{51}$$

For $st \neq 0$ the harmonic matrices do not commute. The resulting holomorphic bundle is locally free with fixed determinant and a three-step filtration whose successive quotients are A.

Proof. For the displayed a, the only nonzero product in $a \wedge a$ is $st(\alpha \wedge \beta) \otimes E_{13}$. Indeed $E_{12}E_{23} = E_{13}$, whereas the reverse product, both squares, and every product involving the correction E_{13} vanish. Thus the displayed form solves the full equation $a = x - d^* G_2(a \wedge a)$ exactly. Contraction uniqueness identifies it with the canonical reconstruction, and Lemma 6.2 gives its harmonic obstruction zero. The full Kuranishi construction gives actual integrability.

Strict upper triangularity preserves the standard flag, with diagonal quotients A. The integrable smooth Dolbeault realization is locally free, and trace zero fixes the determinant. Finally $[sE_{12}, tE_{23}] = stE_{13} \neq 0$. A nonzero commutator cannot become zero under simultaneous conjugation, so this family is outside every conjugate of a commuting or diagonal tuple. □

These bundles are semistable, being successive extensions of equal-slope stable bundles. This does not assert existence of an HE metric on a nonsplit member. The example solves the complete recursion on a particular noncommuting family; it does not give all matrix obstruction equations.

6.3. **Changing stabilizers and separated scales.** Write $u = |s|^2$, $v = |t|^2$. On $\mathfrak{sl}_3(\mathbb{C})$ with Hilbert–Schmidt mass, the complex commutator form is

$$q_{s,t}(\xi) = u\|[\xi, E_{12}]\|^2_{\mathrm{HS}} + v\|[\xi, E_{23}]\|^2_{\mathrm{HS}}. \tag{52}$$

For the geometric mass on $A \otimes \mathbb{C}^3$, the eigenvalues below are divided by $2\,\mathrm{Vol}(\omega_0)$.

Proposition 6.4. *The eight eigenvalues of the complex Hermitian form (52), listed without sorting, are*

$$\begin{gathered} 0,\ v,\ u,\ 2u,\ 2v,\ u+v, \\ u+v-\sqrt{u^2-uv+v^2}, \qquad u+v+\sqrt{u^2-uv+v^2}. \end{gathered} \tag{53}$$

Its complex kernel has dimension one when $uv > 0$, four when exactly one weight vanishes, and eight at the origin. The real Hermitian restriction is discussed below. For $s = \varepsilon\delta$, $t = \varepsilon$, with positive $\varepsilon, \delta \to 0$, three positive eigenvalues have order $\varepsilon^2\delta^2$ and four have order ε^2.

Proof. The orthonormal off-diagonal units $E_{13}, E_{12}, E_{23}, E_{21}, E_{32}, E_{31}$ are eigenvectors with eigenvalues $0, v, u, 2u, 2v, u+v$, respectively. The identity $[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}$ shows that their images in the direct sum of the two target matrix spaces are orthogonal to each other and to the images of diagonal matrices. On $\xi = \operatorname{diag}(d_1, d_2, d_3)$ with $d_1 + d_2 + d_3 = 0$, the form is $u|d_1 - d_2|^2 + v|d_2 - d_3|^2$, with mass $\sum |d_i|^2$. Its eigenvalues on the trace-free plane are the two roots other than the constant-vector root of the weighted path matrix

$$\begin{pmatrix} u & -u & 0 \\ -u & u+v & -v \\ 0 & -v & v \end{pmatrix}.$$

Their sum is $2(u+v)$ and product $3uv$, which gives (53), including the limiting zero weights. For $u/v = \delta^2 \to 0$, the smaller root is

$$\frac{3uv}{u+v+\sqrt{u^2-uv+v^2}} = \tfrac{3}{2}u(1+O(u/v)),$$

and the larger is $2v + O(u)$. This proves both scale counts and the kernel assertions. □

These are complex commutator eigenvalues, not automatically HYM relaxation rates at a nonbalanced point. Their kernel dimensions also differ from those on the real Hermitian algebra. For $uv > 0$ the complex kernel is $\mathbb{C}E_{13}$ and contains no nonzero Hermitian element. On a nonzero one-arrow axis the trace-free Hermitian kernel has real dimension one; at the origin it has dimension eight. Thus the complex dimensions are $1, 4, 8$, whereas the real Hermitian dimensions are $0, 1, 8$.

The same fixed smooth background therefore contains actual noncommuting integrable points with changing stabilizers and arbitrarily separated positive finite scales. The general theorem retains every finite direction throughout this family. For any associated HYM trajectory satisfying its confinement hypothesis, the selected exact graph flow, C^1 initial phase and exponential relative error have the same uniform constants. Determining confinement and the algebraic asymptotics of that corrected finite motion are further dynamical questions.

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